Experiment - 1

Setting up a local area Network

Aim:

To design setup and configure a local area network connecting 4 computers through a network switch assign static IP address to all devices verify connectivity using the ping command and demonstrate data sharing across the network.

IP configuration table:

Device	IP address	subnet mask	Connected to
PC0	125.1.1.0	255.0.0.0	switch Fa0/1
PC1	125.1.1.1	255.0.0.0	switch Fa0/2
PC2	125.1.1.2	255.0.0.0	switch Fa0/3

Implementation:

Step 1: Open Cisco packet tracer and Place switch 4PCs and one server in the workspace.

Step 2: Connect all PC's and the sever to the switch using copper straight Through cables.

Step 3: Configure all IP addresses of all PCs and the server so that they belong to same network.

Step 4: Verify that all cable connection are active

Steps: Test network connectivity by opening the command prompt on a PC using ping command to communicate with another PC or server.

Step 8: Switch from realtime mode to simulation mode in packet traces.

Step 9: Send a ping or web request from a Pc to the server and use the capture forward or auto capture / play option to observe the packet traveling through the switch.

Step 10: Verify that successfully communication takes places btw the Pc's and the server, demonstrating proper LAN operation and data sharing.

Result: A LAN was successfully established using switch and a server, communication btw devices as verified through ping and web access and packet flow was observed successfully in simulation mode.